

Assignment Spring 2025 Report

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Subject: COMP7024 - Programming for Data Science

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Declaration

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Project Assumptions:

The analysis will use the provided CSV files by the assignment. For a comprehensive analysis, the PG1 and PG2 files will be combined into two one data frames for each type of dataset (`patients`, `encounters`, `conditions`). All data files will be included in the repository. The analysis is performed using R within an R Markdown document, which allows for reproducible research by combining code, results, and narrative. As per the project context, all patient ages will be calculated relative to the date **October 13, 2025**. The code will use the column names as they appear in the CSV files (e.g., `Id`, `BIRTHDATE`, `COUNTY`, `PATIENT`, `DESCRIPTION`).

The code in this report have been ensured to run without errors. The codes are well-commented and separated into sections for clarity. All warning and formatting issues have been addressed to ensure a clean and professional presentation.

We first needs to loads all packages required for the assignment, including data manipulation, date calculations, and plot arrangement, and other utilities.

```
knitr::opts_chunk$set(echo = TRUE)

library(tidyverse)
library(lubridate)
library(patchwork)
library(kableExtra)
```

Task 1

The goal is to analyze and visualize the distribution of patients based on two criteria: 1. The geographic distribution of patients diagnosed with COVID-19 (confirmed or suspected) across different counties. 2. The demographic distribution of the entire patient population across four age groups: 0-18, 19-35, 36-50, and 51+.

Data preparation We then handles the loading and preparation of the data. The script explicitly loads the four CSV files and combines them using `rbind()` so that the two patient files are merged as one, and the two condition files are also merged as one.

```
patients_pg1 <- read.csv("patientsPG1.csv")
patients_pg2 <- read.csv("patientsPG2.csv")

patients <- rbind(patients_pg1, patients_pg2)

conditions_pg1 <- read.csv("conditionsPG1.csv")
conditions_pg2 <- read.csv("conditionsPG2.csv")

conditions <- rbind(conditions_pg1, conditions_pg2)
```

Now, we perform the analysis using the combined data frames.

The script first isolates the patient IDs associated with a “COVID” diagnosis and then uses those IDs to filter the main patient list, allowing for an accurate count by county. The script also uses `mutate()` to add new columns. Ages are calculated precisely using `lubridate`, and the `cut()` function is used to efficiently categorize all patients into the defined age bins.

```

# --- Distribution of COVID Patients by County ---
covid_patient_ids <- conditions %>%
  filter(grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  distinct(PATIENT) %>%
  pull(PATIENT)

covid_by_county <- patients %>%
  filter(Id %in% covid_patient_ids) %>%
  count(COUNTY, name = "patient_count") %>%
  arrange(desc(patient_count))

# --- Distribution of Patients by Age Group ---
today <- as.Date("2025-10-13")

patients_by_age_group <- patients %>%
  mutate(
    BirthDate = as.Date(BIRTHDATE),
    Age = floor(time_length(interval(BirthDate, today), "years")),
    `Age Group` = cut(Age,
                      breaks = c(-1, 18, 35, 50, Inf),
                      labels = c("0-18", "19-35", "36-50", "51+"),
                      right = TRUE)
  ) %>%
  count(`Age Group`, name = "patient_count")

```

Visualization After preparing the data, we create two bar plots using `ggplot2`. The first plot visualizes the distribution of COVID-19 patients by county, while the second plot shows the distribution of all patients by age group. The plots are then arranged side-by-side using the `patchwork` library for better comparison.

Both distributions were visualized using bar plots instead of histograms, as bar plots are more appropriate for categorical data like counties and age groups.

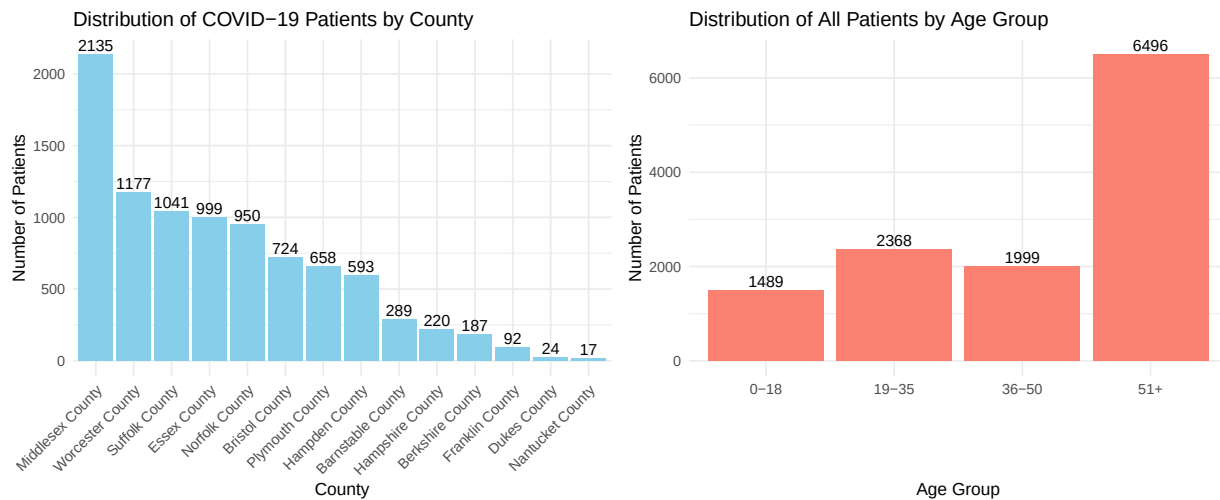
```

# Plot 1: COVID Patients by County
plot1 <- ggplot(covid_by_county,
               aes(x = reorder(COUNTY, -patient_count), y = patient_count)) +
  geom_col(fill = "skyblue") +
  geom_text(aes(label = patient_count), vjust = -0.3) +
  labs(title = "Distribution of COVID-19 Patients by County",
       x = "County",
       y = "Number of Patients") +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Plot 2: All Patients by Age Group
plot2 <- ggplot(patients_by_age_group,
               aes(x = `Age Group`, y = patient_count)) +
  geom_col(fill = "salmon") +
  geom_text(aes(label = patient_count), vjust = -0.3) +
  labs(title = "Distribution of All Patients by Age Group",
       x = "Age Group",
       y = "Number of Patients") +
  theme_minimal(base_size = 12)

```

```
# Arrange the two plots side-by-side using the patchwork library.
plot1 + plot2
```



The bar chart on the left shows the absolute number of patients with a COVID-19 diagnosis across the different counties in Massachusetts. There is a significant geographic concentration of COVID-19 cases. Middlesex County stands out as the area with the highest number of patients, totaling 2,135. This figure is nearly double that of the next highest county, Worcester.

The top eight counties collectively account for the vast majority of the COVID-19 cases in this dataset. This suggests that these are major hotspots. This information indicates that medical resources and public health interventions may need to be concentrated in these areas to effectively manage and contain the spread of the virus.

The bar chart on the right provides a demographic snapshot of the entire patient population. The patient population is heavily skewed towards older adults. The 51+ age group is by far the largest demographic, with 6,496 patients. This is more than the other three age groups combined. This distribution strongly suggests that older adults are more likely to seek medical care or have more health issues requiring attention. There is a surprising spike when people reach the age of 51 compared to the 36-50 age group. This could be due to a variety of factors, including the onset of age-related health conditions that become more prevalent after 50 or that people are more proactive about their health as they age. This information means that more pressure may be placed on healthcare services that cater to older adults, including chronic disease management and preventive care.

Task 2

1. Identify the top 5 most common conditions for all COVID-19 patients, excluding conditions that are directly related to the COVID-19 diagnosis itself.
2. Investigate if these common co-occurring conditions differ by gender.
3. Present the top 10 conditions for male and female patients separately in a formatted table for comparison and provide an interpretation of the findings.

Data preparation For task 2, we reuse the `covid_patient_ids` vector created in Task 1 to filter the `conditions` data frame. This ensures that we are only analyzing conditions for patients who have had a COVID-19 diagnosis.

The code block below uses a `dplyr` pipe to chain operations. It `filter()`s for our COVID patient cohort, then uses `filter()` with a `!` (negation) to *remove* all COVID-related diagnoses. Finally, `count()` tallies the remaining conditions, and `top_n(5)` selects the five most frequent ones. A crucial step here is the `inner_join()`, which merges the two data frames, allowing us to analyze conditions and demographics together. The logic is then repeated twice, adding a `filter()` for `GENDER == 'M'` and `GENDER == 'F'` respectively to create two separate top-10 lists.

```
# --- Find Top 5 Most Common Conditions (Overall) ---
top_5_conditions_overall <- conditions %>%
  filter(PATIENT %in% covid_patient_ids) %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  top_n(5, count)

print(top_5_conditions_overall)
```

```
##              DESCRIPTION count
## 1              Fever (finding) 8083
## 2              Cough (finding) 6202
## 3      Loss of taste (finding) 4711
## 4              Fatigue (finding) 3516
## 5 Body mass index 30+ - obesity (finding) 3490
```

```
# --- Analyze Top 10 Conditions by Gender ---

conditions_with_gender <- conditions %>%
  inner_join(patients, by = c("PATIENT" = "Id"))

top_10_male <- conditions_with_gender %>%
  filter(PATIENT %in% covid_patient_ids, GENDER == 'M') %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  head(10)

top_10_female <- conditions_with_gender %>%
  filter(PATIENT %in% covid_patient_ids, GENDER == 'F') %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  head(10)
```

The code block above prints the overall top 5 health issues experienced by COVID-19 patients in this dataset. Fever is the most common condition with a count of 8083, followed by Cough, Loss of taste, Fatigue, and obesity with a body mass index over 30. Identifying these conditions is crucial for patient management, as it points to which other conditions doctors should monitor closely in a COVID-19 patient.

Visualization The next code block produces a side-by-side comparison of the top 10 conditions for male and female patients.

The data are first prepared by renaming columns for clarity. The `cbind()` function is then used to combine the two data frames into one for side-by-side comparison. Finally, `kbl()` and `kable_styling()` are used to format the table with a caption and styling options.

Table 1: Top 10 Most Common Non-COVID Conditions in COVID-19 Patients by Gender

Male Patients		Female Patients	
Male Condition	M_Count	Female Condition	F_Count
Fever (finding)	3833	Fever (finding)	4250
Cough (finding)	2936	Cough (finding)	3266
Loss of taste (finding)	2271	Loss of taste (finding)	2440
Fatigue (finding)	1695	Body mass index 30+ - obesity (finding)	1918
Body mass index 30+ - obesity (finding)	1572	Fatigue (finding)	1821
Sputum finding (finding)	1406	Miscarriage in first trimester	1580
Anemia (disorder)	1387	Sputum finding (finding)	1564
Prediabetes	1302	Prediabetes	1377
Hypertension	1084	Hypertension	1183
Hypoxemia (disorder)	865	Normal pregnancy	1063

```
# --- Present the Gender Comparison in a Table ---

male_table_data <-
  top_10_male %>% rename(`Male Condition` = DESCRIPTION, `M_Count` = count)
female_table_data <-
  top_10_female %>% rename(`Female Condition` = DESCRIPTION, `F_Count` = count)

comparison_table <- cbind(male_table_data, female_table_data)

comparison_table %>%
  kbl(caption = "Top 10 Most Common Non-COVID
    Conditions in COVID-19 Patients by Gender") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) %>%
  add_header_above(c("Male Patients" = 2, "Female Patients" = 2))
```

According to the table above, eight of the top 10 conditions are shared between male and female patients. Classic viral symptoms such as Fever, Cough, Loss of taste, and Fatigue rank in the top 5 for both genders. This is consistent with the well-documented presentation of COVID-19 and confirms the symptomatic overlap between men and women.

The presence of Obesity (BMI 30+), Prediabetes, and Hypertension on both lists is highly significant. These conditions are widely recognized as major risk factors that increase the likelihood of severe complications from COVID-19. Their high frequency underscores the role of metabolic and cardiovascular health in disease outcomes for all patients.

For male, the unique conditions points towards severe respiratory issues. The appearance of Hypoxemia, or low blood oxygen levels, is a direct indicator of this concern. While anemia can be associated with chronic inflammation and more severe systemic illness, it may reflect that men were experiencing an inflammatory response to COVID-19.

For female, the unique conditions points towards pregnancy-related issues. The high presence of miscarriage is alarming. This finding emphasizes the need for special care and monitoring of pregnant patients with COVID-19, as they may face unique risks and complications. However, there is also the presence of normal pregnancy, which suggests that not all pregnant patients with COVID-19 experience adverse outcomes, and that the dataset highlights that pregnant women are more likely to contract COVID-19 compared to other demographics.

Task 3

1. Analyze factors that may influence the hospitalization rate for patients with COVID-19 focusing on Age Group and Gender.

We will reuse the `patients` and `conditions` data frames created in Task 1. However, since the assignment provides two separate datasets (PG1 and PG2), we will perform the analysis separately for each dataset to see if there are any differences in hospitalization factors between them.

Data preparation Following the DRY principle, we encapsulate the analysis logic into a function called `analyze_hospitalization_factors()`. This function takes in the patient, condition, and encounter data frames along with a dataset name for labeling purposes. It performs the necessary joins and calculations to determine hospitalization rates by age group and gender.

Inside the function we first filter the encounters to include only those related to COVID-19 patients. We then join this filtered data with the patient data to get demographic information. New columns for hospitalization status, age, and age group are created using `mutate()`. The function then calculates hospitalization rates by age group and gender separately, returning the results as a list.

```
encounters_pg1 <- read.csv("encountersPG1.csv")
encounters_pg2 <- read.csv("encountersPG2.csv")

analyze <- function(patients_df, conditions_df, encounters_df, dataset_name) {
  covid_patient_ids <- conditions_df %>%
    filter(grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
    distinct(PATIENT) %>%
    pull(PATIENT)

  # Join encounters with patient data for the COVID cohort
  analysis_data <- encounters_df %>%
    filter(PATIENT %in% covid_patient_ids) %>%
    inner_join(patients_df, by = c("PATIENT" = "Id")) %>%
    mutate(
      Hospitalization =
        if_else(
          ENCOUNTERCLASS == "inpatient", "Hospitalized", "Outpatient"),
      BirthDate = as.Date(BIRTHDATE),
      Age = floor(
        time_length(interval(BirthDate,
                             as.Date("2025-10-15")), "years")),
      `Age Group` = cut(Age,
                        breaks = c(-1, 18, 35, 50, Inf),
                        labels = c("0-18", "19-35", "36-50", "51+"))
    ) %>%
    filter(!is.na(`Age Group`), !is.na(GENDER))

  # --- Analyze by Age Group ---
  age_analysis <- analysis_data %>%
    group_by(`Age Group`, Hospitalization) %>%
    summarise(count = n(), .groups = "drop") %>%
    group_by(`Age Group`) %>%
```

```

mutate(rate = count / sum(count)) %>%
filter(Hospitalization == "Hospitalized")

# --- Analyze by Gender ---
gender_analysis <- analysis_data %>%
  group_by(GENDER, Hospitalization) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(GENDER) %>%
  mutate(rate = count / sum(count)) %>%
  filter(Hospitalization == "Hospitalized")

(list(age = age_analysis,
      gender = gender_analysis,
      name = dataset_name))
}

results_pg1 <- analyze(patients_pg1,
                      conditions_pg1,
                      encounters_pg1, "PG1 Dataset")
results_pg2 <- analyze(patients_pg2,
                      conditions_pg2,
                      encounters_pg2, "PG2 Dataset")

```

Visualization The next code block produces 4 plots to compare hospitalization rates between the two datasets (PG1 and PG2) for both age groups and genders. The plots are arranged in a 2x2 grid for easy comparison.

```

# Plot for Age Group comparison
plot_age_pg1 <- ggplot(results_pg1$age, aes(x = `Age Group`, y = rate)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = scales::percent(rate, accuracy = 0.1)), vjust = -0.3) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Hospitalization Rate by Age Group (PG1)",
       y = "Hospitalization Rate") +
  theme_minimal()

plot_age_pg2 <- ggplot(results_pg2$age, aes(x = `Age Group`, y = rate)) +
  geom_col(fill = "darkorange") +
  geom_text(aes(label = scales::percent(rate, accuracy = 0.1)), vjust = -0.3) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Hospitalization Rate by Age Group (PG2)",
       y = "Hospitalization Rate") +
  theme_minimal()

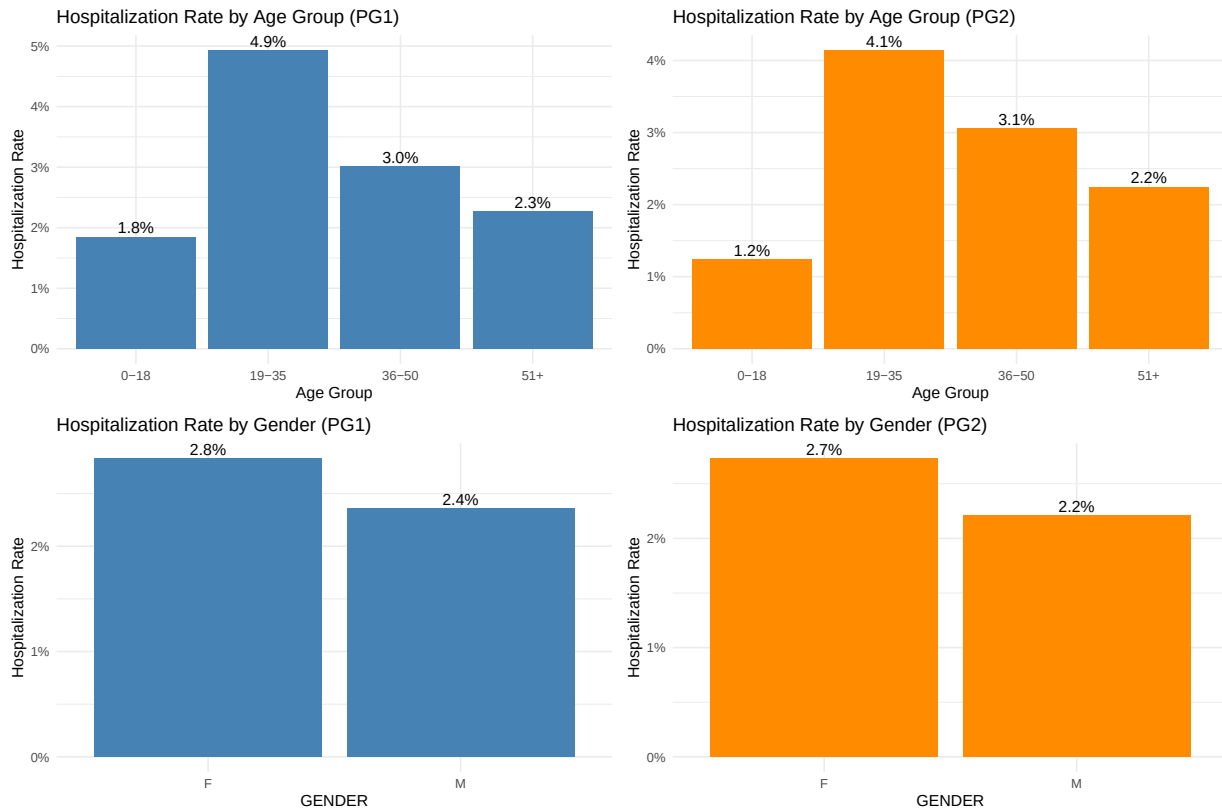
# Plot for Gender comparison
plot_gender_pg1 <- ggplot(results_pg1$gender, aes(x = GENDER, y = rate)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = scales::percent(rate, accuracy = 0.1)), vjust = -0.3) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Hospitalization Rate by Gender (PG1)",
       y = "Hospitalization Rate") +
  theme_minimal()

```



```
plot_gender_pg2 <- ggplot(results_pg2$gender, aes(x = GENDER, y = rate)) +
  geom_col(fill = "darkorange") +
  geom_text(aes(label = scales::percent(rate, accuracy = 0.1)), vjust = -0.3) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Hospitalization Rate by Gender (PG2)",
       y = "Hospitalization Rate") +
  theme_minimal()

(plot_age_pg1 | plot_age_pg2) / (plot_gender_pg1 | plot_gender_pg2)
```



Both PG1 and PG2 datasets show a consistent, similar trend for both age group and gender. For the age group, the charts show a non-linear trend. It is worth noting that despite the age group 51+ having the highest absolute number of COVID-19 patients, they do not have the highest hospitalization rate. Instead, the 19-35 age group has the highest hospitalization rate in both datasets. There can be several reasons for the highest age group not having the highest hospitalization rate. It could be that older adults are more hesitant to seek hospital care unless absolutely necessary, or that they have better access to outpatient care and management of their conditions. The 19-35 age group may have more severe cases that require hospitalization, or they may be more likely to seek hospital care when they experience symptoms. The youngest age group has the lowest hospitalization rate.

There is also not much difference between PG1 dataset and PG2 dataset when it comes to hospitalization rate by gender. The comparison shows that females have a higher hospitalization rate than males. It is highly likely that pregnancy is a contributing factor to this trend. Pregnant women may be more likely to be hospitalized for COVID-19 due to concerns about the health of both the mother and the baby.

Task 4

1. Investigate the characteristics of COVID-19 patients who recover compared to those who do not.
2. Identify factors that impact recovery outcomes and to assess if these factors are consistent across the two separate datasets (PG1 and PG2).

Data preparation To handle the complexity of this task, we encapsulate the analysis logic into a function called `analyze_recovery()`. This function takes in the patient and condition data frames along with a dataset name for labeling purposes. It performs the necessary joins and calculations to determine recovery status and analyze various factors. Some notable steps in the function include:

- Identifying COVID-19 conditions and defining recovery status based on the presence of a STOP date.
- Calculating age and categorizing patients into age groups.
- Analyzing recovery rates by age group and gender
- Analyzing median recovery time
- Identifying top conditions for recovered and non-recovered patients.

```
analyze_recovery <- function(patients_df, conditions_df, dataset_name) {  
  # --- Identify COVID Conditions and Define Recovery Status ---  
  covid_conditions <- conditions_df %>%  
    filter(grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%  
    mutate(  
      START = as.Date(START),  
      STOP = as.Date(STOP)  
    )  
  
  patient_recovery_status <- covid_conditions %>%  
    group_by(PATIENT) %>%  
    summarise(  
      RecoveryStatus = if_else(all(!is.na(STOP)), "Recovered", "Not Recovered"),  
      IllnessDuration = if_else(RecoveryStatus == "Recovered",  
                               as.numeric(max(STOP, na.rm = TRUE) - min(START, na.rm = TRUE)),  
                               NA_real_)  
    )  
  
  # --- Prepare the Main Analysis Dataset ---  
  analysis_data <- patient_recovery_status %>%  
    inner_join(patients_df, by = c("PATIENT" = "Id")) %>%  
    mutate(  
      BirthDate = as.Date(BIRTHDATE),  
      Age = floor(time_length(interval(BirthDate,  
                                       as.Date("2025-10-15"))), "years"),  
      `Age Group` = cut(Age,  
                        breaks = c(-1, 18, 35, 50, Inf),  
                        labels = c("0-18", "19-35", "36-50", "51+"))  
    ) %>%  
    filter(!is.na(`Age Group`), GENDER %in% c("M", "F"))  
  
  # --- Analyze Age and Gender Recovery Rates ---  
  age_recovery_rate <- analysis_data %>%  
    group_by(`Age Group`, RecoveryStatus) %>%  
    summarise(count = n(), .groups = 'drop') %>%  
    group_by(`Age Group`) %>%
```

```

mutate(rate = count / sum(count)) %>%
filter(RecoveryStatus == 'Recovered')

gender_recovery_rate <- analysis_data %>%
  group_by(GENDER, RecoveryStatus) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(GENDER) %>%
  mutate(rate = count / sum(count)) %>%
  filter(RecoveryStatus == "Recovered")

# --- Analyze Timeline (Median Recovery Time) ---
median_recovery_time <- analysis_data %>%
  filter(RecoveryStatus == "Recovered") %>%
  summarise(MedianDays = median(IllnessDuration, na.rm = TRUE))

# --- Analyze Top Conditions for Each Group ---
all_patient_conditions <- conditions_df %>%
  inner_join(patient_recovery_status, by = "PATIENT")

top_conditions_recovered <- all_patient_conditions %>%
  filter(RecoveryStatus == "Recovered",
         !grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "Count") %>%
  head(5) %>%
  rename(`Recovered Condition` = DESCRIPTION)

top_conditions_not_recovered <- all_patient_conditions %>%
  filter(RecoveryStatus == "Not Recovered",
         !grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "Count") %>%
  head(5) %>%
  rename(`Not Recovered Condition` = DESCRIPTION, `NR_Count` = Count)

df_recovered <-
  top_conditions_recovered %>% mutate(row_num = row_number())
df_not_recovered <-
  top_conditions_not_recovered %>% mutate(row_num = row_number())
symptom_comparison <-
  full_join(df_recovered, df_not_recovered, by = "row_num") %>%
  select(-row_num)

list(
  age_rate = age_recovery_rate,
  gender_rate = gender_recovery_rate,
  timeline = median_recovery_time,
  symptoms = symptom_comparison,
  name = dataset_name
)
}

```

Visualization We then call the `analyze_recovery()` function for both datasets and store the results. We also produce 4 plots to compare recovery rates between the two datasets (PG1 and PG2) for both age groups and genders similar to Task 3.

```

results_pg1 <- analyze_recovery(patients_pg1, conditions_pg1, "PG1 Dataset")
results_pg2 <- analyze_recovery(patients_pg2, conditions_pg2, "PG2 Dataset")

plot_age_pg1 <- ggplot(results_pg1$age_rate, aes(x = `Age Group`, y = rate)) +
  geom_col(fill = "forestgreen") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Recovery Rate by Age Group (PG1)", y = "Recovery Rate") +
  theme_minimal()

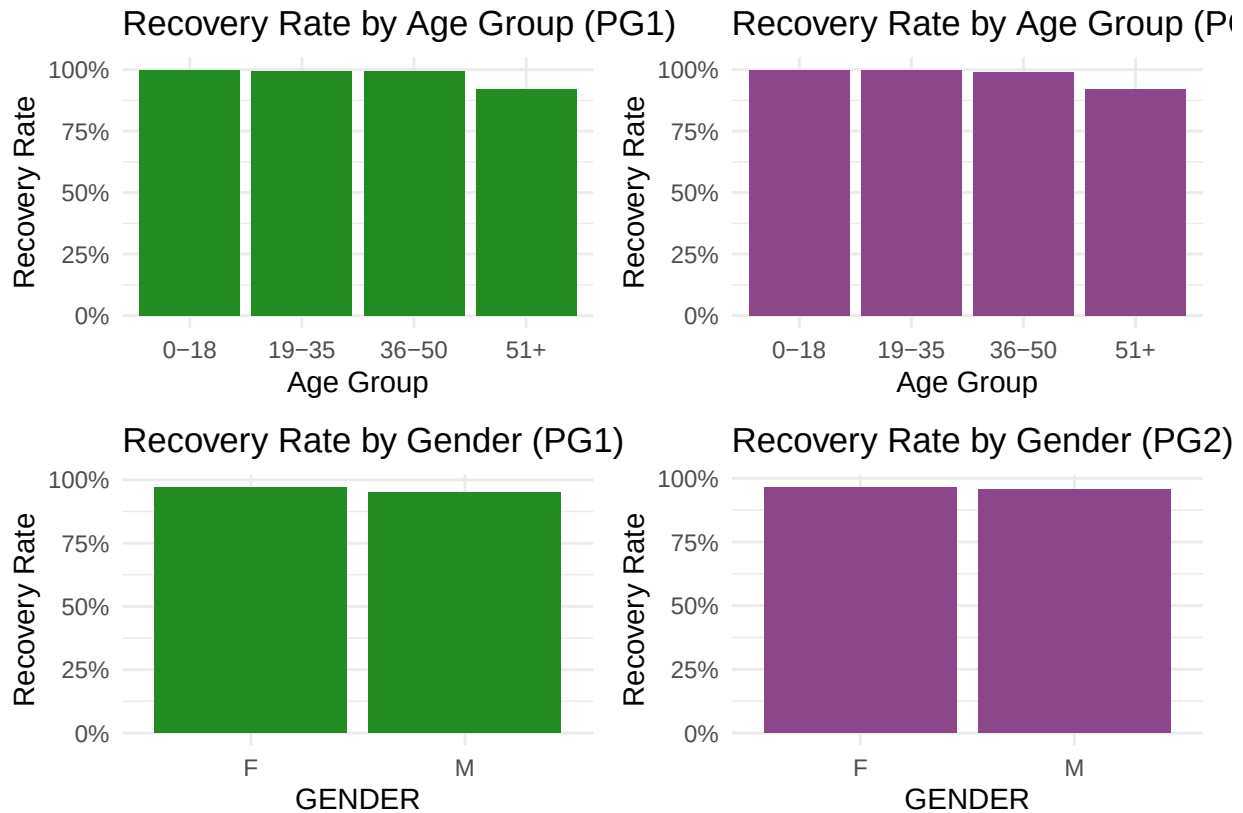
plot_age_pg2 <- ggplot(results_pg2$age_rate, aes(x = `Age Group`, y = rate)) +
  geom_col(fill = "orchid4") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Recovery Rate by Age Group (PG2)", y = "Recovery Rate") +
  theme_minimal()

plot_gender_pg1 <- ggplot(results_pg1$gender_rate, aes(x = GENDER, y = rate)) +
  geom_col(fill = "forestgreen") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Recovery Rate by Gender (PG1)", y = "Recovery Rate") +
  theme_minimal()

plot_gender_pg2 <- ggplot(results_pg2$gender_rate, aes(x = GENDER, y = rate)) +
  geom_col(fill = "orchid4") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Recovery Rate by Gender (PG2)", y = "Recovery Rate") +
  theme_minimal()

(plot_age_pg1 | plot_age_pg2) / (plot_gender_pg1 | plot_gender_pg2)

```



The charts above shows that there is an exceptionally high recovery rate across all age groups in both datasets. The 0-18 age group has the highest recovery rate, followed closely by the 19-35 and 36-50 age groups. The 51+ age group has the lowest recovery rate, this is consistent with the general understanding that older adults are more vulnerable to severe outcomes from COVID-19. Age appears to be a significant factor in recovery, with younger patients recovering at higher rates.

Furthermore, the recovery rates of female patients are higher than those of male. This could be due to a variety of biological, behavioral, and social factors. For instance, there is some evidence to suggest that hormonal differences may influence immune responses to infections. Additionally, women may be more proactive in seeking medical care and adhering to treatment regimens.

Both datasets show similar trends in both of these comparisons.

```
cat("PG1 Dataset:", results_pg1$timeline$MedianDays, "days\n")
```

```
## PG1 Dataset: 22 days
```

```
cat("PG2 Dataset:", results_pg2$timeline$MedianDays, "days\n\n")
```

```
## PG2 Dataset: 22 days
```

Both datasets show a median recovery time of 22 days for patients who recover from COVID-19. This consistency suggests that, despite any differences in patient demographics or healthcare settings between the two datasets, the typical duration of illness remains similar.

Table 2: Top Conditions for Recovered vs. Not Recovered Patients (PG1)

Recovered Condition	Count	Not Recovered Condition	NR_Count
Fever (finding)	4423	Hypoxemia (disorder)	195
Cough (finding)	3400	Pneumonia (disorder)	195
Loss of taste (finding)	2584	Respiratory distress (finding)	195
Body mass index 30+ - obesity (finding)	1909	Sepsis caused by virus (disorder)	195
Fatigue (finding)	1906	Acute respiratory failure (disorder)	193

Table 3: Top Conditions for Recovered vs. Not Recovered Patients (PG2)

Recovered Condition	Count	Not Recovered Condition	NR_Count
Fever (finding)	3338	Hypoxemia (disorder)	150
Cough (finding)	2569	Pneumonia (disorder)	150
Loss of taste (finding)	1948	Respiratory distress (finding)	150
Fatigue (finding)	1468	Sepsis caused by virus (disorder)	150
Body mass index 30+ - obesity (finding)	1384	Acute respiratory failure (disorder)	148

```
results_pg1$symptoms %>%
  kbl(caption = "Top Conditions for Recovered vs.
    Not Recovered Patients (PG1)") %>%
  kable_styling(full_width = FALSE, position = "center")
```

```
results_pg2$symptoms %>%
  kbl(caption = "Top Conditions for Recovered vs.
    Not Recovered Patients (PG2)") %>%
  kable_styling(full_width = FALSE, position = "center")
```

Compared to PG2, PG1 shows a higher count for both the recovered conditions and non-recovered conditions. Fever and Cough are the most common conditions among recovered patients in both datasets, indicating that these symptoms are prevalent but not necessarily indicative of poor outcomes. Fatigue and Loss of taste are also common among recovered patients.

PG2 and PG1 shows the exact same rankings for the not recovered conditions. Hypoxemia and Pneumonia are the most common conditions among non-recovered patients in both datasets, highlighting the severity of respiratory complications in COVID-19 outcomes. Acute respiratory failure, Respiratory distress and sepsis caused by virus are also prevalent among non-recovered patients, underscoring the critical nature of these conditions in determining patient outcomes.

This list describes a patient whose body is failing to cope with the viral infection. The conditions point directly to severe lung damage and a systemic inflammatory crisis, which are the primary drivers of mortality and long-term illness from COVID-19. These are critical medical emergencies that require intensive care.

Compared to PG1, Fatigue have a higher count than body mass index over 30 for the recovered conditions in PG2. But aside from that, both datasets show the exact same rankings for the recovered conditions as well.

Compared to the recovered conditions, the not recovered conditions tend to be more severe and life-threatening. This contrast highlights the importance of monitoring and managing respiratory symptoms in COVID-19 patients to improve recovery outcomes. Recovered conditions tend to be more general symptoms of viral infections.

Recovery is associated with experiencing the symptoms of the virus, while non-recovery is defined by the development of severe complications.